

SWARAJ SAMBHAJI YADAV

Data Engineer

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SUMMARY

Data Engineer with 4 years of experience specializing in designing scalable data solutions that drive operational efficiency. Expert in building high-performance ETL pipelines, implementing robust data governance, and leveraging advanced analytics to optimize business processes. Proven track record of reducing costs, enhancing data integrity, and enabling data-driven decision-making through innovative technical interventions.

SKILLS

Methodologies: SDLC, Agile, Waterfall

Programming Languages: Python (Pandas, Numpy, Matplotlib, seaborn), SQL, R, Java

Machine Learning Libraries: TensorFlow, Pytorch, Keras, SciPy, NumPy, Scikit-learn

Big Data Technologies: Apache Airflow, Hadoop, Hive, HDFS, Apache Flink, Sqoop, Spark, Kafka

Data Base: MySQL, PostgreSQL, Microsoft SQL Server, Oracle, RDS, NoSQL, MongoDB, Dynamo DB

Data Visualization Tools: Tableau, Power BI

Data Preprocessing: Data Cleaning, Data Modelling, A/B Testing, Statistical Analysis, Descriptive Analytics, Data Mining

ETL and Data Analytics Platforms: SSIS, SSAS, Alteryx, Talend, Informatica, DBT (data build tool)

Cloud Platforms: AWS (S3, Glue, RedShift), Azure (Synapse, Data Factory), GCP (BigQuery), Snowflake, Data Bricks

Containerization: Docker, Kubernetes

Version Control: Git, GitHub

Infrastructure as Code: Terraform, Jenkins, CircleCI

Soft Skills: Critical Thinking, Strong Analytical & Problem-Solving skills, Project Management, Attention to detail, Effective Communication, Collaboration with Teams

EDUCATION

Master of Science in Data Science | Rochester Institute of Technology, Rochester, NY

May 2024

Bachelor of Engineering in Computer Engineering | University of Mumbai, India

November 2020

WORK EXPERIENCE

Data Engineer | BNY Mellon | New York

May 2023 - Current

- Designed, built, and maintained high-volume ETL pipelines using Apache Airflow, Apache Spark, and AWS Glue, processing financial data from trading systems and third-party providers, improving data processing speeds, enabling faster reporting and enhancing data-driven decision-making for key stakeholders.
- Developed scalable data models using SQL, MySQL, and Snowflake, consolidating over 10 terabytes of structured and unstructured data, streamlined compliance monitoring and real-time risk analysis, reducing reporting cycles and minimizing operational delays.
- Implemented robust data governance policies using Terraform for infrastructure automation and AWS IAM for data access control, ensuring adherence to industry standards like PCI DSS, enhanced data security, reducing potential breach risks by 40% through proactive monitoring and automated compliance checks.
- Built fraud detection models using Python libraries (Pandas, NumPy, Scikit-learn) and deployed them on AWS SageMaker, detecting and flagging 10 fraudulent transactions monthly, reducing manual investigation time by 35%.
- Conducted comprehensive data validation, statistical analysis, and data cleaning using Python (Pandas, SciPy), enhanced regulatory reporting accuracy by reducing data-related audit issues through automated data consistency checks and statistical anomaly detection.
- Collaborated with data scientists, business analysts, and compliance teams to align data solutions with business goals, create interactive dashboards using Power BI, tracking key financial metrics such as revenue, transaction volumes, and risk indicators. These dashboards improved executive decision-making efficiency through real-time business insights.

Data Engineer | Trigent Software | India

February 2019 - July 2021

- Developed scalable ETL pipelines using Python (Pandas, NumPy), SQL, and Apache Airflow for batch and real-time data ingestion, leveraged AWS Glue and Kinesis to integrate diverse data sources into cloud platforms like Redshift and Databricks, improving data availability for analytics.
- Automated CI/CD workflows for data integration using Jenkins, Terraform, and CircleCI, ensuring consistent pipeline deployments and reducing deployment times by 50%, utilized version control systems (Git, GitHub) to streamline collaboration and enhance code reliability across teams.
- Optimized cloud data storage and processing efficiency using Hadoop for distributed storage and Spark for parallel computations, achieving a 30% reduction in storage costs, designed and applied query-optimized data models in PostgreSQL and BigQuery, improving analytics performance.
- Implemented data quality frameworks using Python and Apache Airflow to validate, clean, and preprocess large datasets, reduced ETL errors, enabling reliable downstream analytics and ensuring data integrity for business-critical decisions.
- Proactively monitored pipeline health and resolved bottlenecks using Datadog and Kubernetes, maintaining 99.9% system uptime across production environments, deployed containerized workflows with Docker, improving scalability and consistency.
- Collaborated with cross-functional teams to define data models, conduct descriptive analytics, and support data-driven experiments through A/B testing infrastructure, contributed to a 25% productivity improvement by enabling faster decision-making and streamlined data processes.

PROJECT

Capstone Project | Rochester, NY (Co-evolving RNNs and their Hyperparameters with SHO)

Created an advanced algorithm for hyperparameter tuning in the Evolutionary Exploration of Augmented Memory Model (EXAMM), boosting RNN performance by 15%. Dispatched the (SHO) algorithm, significantly improving the efficiency and accuracy of RNN training. Published findings in GECCO 2023 showcasing a 15% performance improvement in neuro-evolution of RNNs due to the tuning algorithm.